

Medetomidine in the Illicit Drug Supply

Lynch MJ, Pizon AF, Yealy DM. Ann Emerg Med. 2026;87:709-716.



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Clinical question

How should clinicians recognize and manage medetomidine-associated intoxication and withdrawal when it appears with illicit opioids?

What the paper says

- Medetomidine is a veterinary alpha-2 agonist increasingly detected with illicit fentanyl.
- Naloxone may restore ventilation while medetomidine sedation persists.
- Withdrawal can begin within 4-6 hours and progress rapidly with severe autonomic instability, vomiting, agitation, delirium, and hyperthermia.
- The authors describe early antiemetic therapy, oral/transdermal alpha-2 agonists, and IV dexmedetomidine for refractory cases.

Evidence guardrails

- Narrative review/clinical concepts paper, not a randomized treatment trial or validated guideline.
- Medetomidine-specific evidence is largely case reports, case series, pharmacology, and expert experience.
- Illicit exposures are polysubstance exposures; attribution and testing are limited.

SAFETY: An unusually early, severe withdrawal picture should not automatically be treated as opioid withdrawal based only on COWS. Consider medetomidine or another adulterant and obtain toxicology/poison-center support.

Clinical pattern

Intoxication

- Prolonged sedation, miosis, bradycardia, initial hypertension followed by hypotension.
- Naloxone remains indicated for opioid respiratory depression but does not reverse medetomidine.
- Supportive care, aspiration precautions, fluids, and airway protection are central.

Withdrawal

- Anxiety, tremor, diaphoresis, nausea/vomiting, tachycardia, hypertension, agitation, delirium, and hyperthermia.
- Possible hypokalemia, acidosis, QTc prolongation, cardiomyopathy, and encephalopathy.
- Standard opioid or GABA-agonist treatment may have little effect.

Proposed management framework

Recognize / monitor	Stabilize symptoms	Address OUD separately
• Clinical diagnosis; monitor airway, breathing, circulation, ECG, electrolytes, mental status, and trajectory. • Early dopamine-antagonist antiemetic therapy; oral clonidine plus scheduled guanfacine; transdermal clonidine for tapering; escalate to IV dexmedetomidine when refractory. • Avoid premature buprenorphine based only on COWS; consider carefully titrated full-agonist support and micro-induction strategies when appropriate; monitor QTc with methadone.		

PROFESSIONAL CONTEXT: CDC materials emphasize that atipamezole is not an established human reversal strategy and may precipitate dangerous withdrawal. CDC and UPMC education reinforce toxicology consultation and the evolving evidence base.

Clinical implications for ASAM 3.7 / 3.5 practice

- Ask about changing drug-supply effects when a patient has prolonged sedation after naloxone or unusually early severe autonomic withdrawal.
- Do not use “opioid withdrawal” as the only explanatory label for early severe symptoms.
- Create an escalation pathway to emergency medicine, medical toxicology, poison center, ICU, pharmacy, addiction medicine, and public health.
- Separate acute stabilization from eventual MOUD initiation; reassess physiology, timing, opioid exposure, and withdrawal phenotype.
- Provide return precautions for delayed vomiting, diaphoresis, tremor, agitation, chest pain, or vital-sign abnormalities when discharge is appropriate.

Discussion prompts

1. Which features best distinguish medetomidine withdrawal from opioid, benzodiazepine, stimulant, or xylazine withdrawal?
2. When is COWS insufficient for deciding buprenorphine timing?
3. Which parts of the proposed algorithm are feasible in our setting?
4. How should suspected exposure be documented when testing is unavailable?
5. What harm-reduction and discharge instructions are needed as the drug supply changes?

Sources

Primary article: <https://doi.org/10.1016/j.annemergmed.2025.12.004>

Additional professional context: CDC clinician webinar and Q&A; CDC Health Advisory HAN00527; PubMed clinical update; UPMC clinical education; American Family Physician commentary.

For journal-club education and clinical discussion. Not individualized medical advice. Follow local emergency, toxicology, pharmacy, and critical-care guidance.